Princess Sumaya University for Technology - King Hussein School for Computing Sciences

CS11313 - **Algorithms Design and Analysis** - **Fall** 2021

Quiz # 2

Name:	**ID:**

Q. Solve the following recurrence equations using the Master Method. If the Master Method does not apply, explain briefly why and then solve the equation using iterative substitution ***or*** the recursion tree method.

A. $T(n) = 8T(\frac{n}{4}) + c, T(1) = c$

Answer. (5 points)

$\log_4 8 = \frac{\log_2 8}{\log_2 4} = \frac{3}{2} = 1.5$

Hence, $n^{\log_a b} = n^{1.5}$.

$c = O(n^{1.5-\epsilon})$ if we pick $\epsilon \leq 1.5$ (we are in Case 1 of the Master Method)

Therefore, $T(n) = \Theta(n^{1.5})$

B. $T(n) = 3T(n-2) + c, \quad T(1) = c, \; T(0) = c$

Answer. (5 points) The Master Method does not apply because the recurrence does not follow the form $T(n) = aT(n/b)$

Recursion Tree Method.

Work at level $i = 3^i c$ Tree height = $\frac{n}{2}$ Total work = $c \times \sum_{i=0}^{\frac{n}{2}} 3^i$

Total work = $c(\frac{3^{\frac{n}{2}+1} - 1}{3-1}) \;=\; \frac{1}{2}c(3 \times \sqrt{3}^n - 1) \;=\; \frac{3}{2}c\sqrt{3}^n - \frac{c}{2} \;=\; \Theta(\sqrt{3}^n)$

Iterative substitution

$T(n) = c + 3T(n-2) = c + 3(c + 3T(n-4))$

$T(n) = c + 3c + 9T(n-4) = c + 3c + 9(c + 3T(n-6))$

$T(n) = c + 3c + 9c + 27T(n-6)$

For some $k < \frac{n}{2}$: $T(n) = 3^0c + 3^1c + 3^2c + ... + 3^{k-1}c + 3^kT(n-2k)$

When $k = \frac{n}{2}$: $T(n) = c \times \sum_{i=0}^{\frac{n}{2}} 3^i$ = $c(3^{\frac{n}{2}+1} - 1) \;=\; 3c\sqrt{3}^n - c \;=\; \Theta(\sqrt{3}^n)$